



ASX Clearing House Electronic Subregister System (CHES) Replacement Project

Project Management Case Study
Analysis

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Background



- **Clearing House Electronic Subregister System (CHES)** has supported Australia's securities clearing and settlement since 1994.
- ASX launched the replacement project in 2016 using **Distributed Ledger Technology (DLT)** to modernise the system.
- Governance and delivery challenges led to the project's failure in 2022.



Project Objectives



Modernise Australia's clearing & settlement infrastructure



Improve transaction efficiency



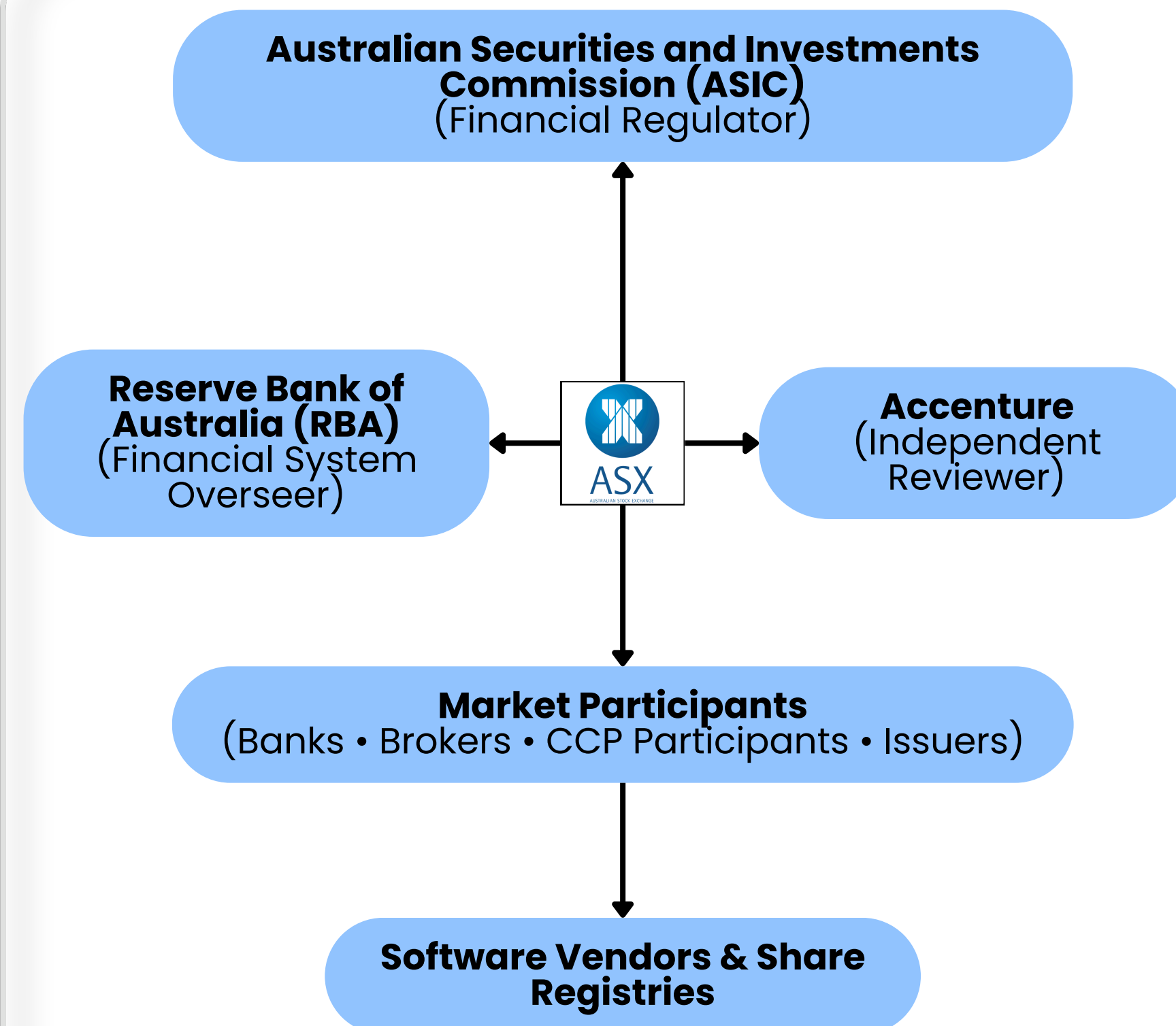
Enhance resilience, security & scalability



Support future innovation and market services



Key Stakeholders





Timeline

- 
2016
 Project initiated
- 
2017 – 2018
 Requirements & solution design
- 
2019 – 2021
 Development, testing & industry consultation
- 
2022 (Nov)
 Project failed (no go-live)



Scope

In Scope

-  Replace CHES with DLT-based platform
-  Improve scalability, resilience, and efficiency
-  Enhance settlement speed and capacity
-  Support ISO 20022 readiness

Out of Scope

-  Replacement of other ASX systems
-  Other clearing & payment systems
-  Non-settlement business processes



Budget (AUD)



Initial Investment
 ~ AUD 75–85 million



Financial Loss
 ~ AUD 245–255 million (write-off)



Financial Impact
 Significant cost escalation and investment write-off



Final Outcome



Project Failed
 November (2022)

Outcomes

- Governance failures
- Delivery delays
- Increased project risks
- Loss of stakeholder confidence
- Original DLT solution abandoned

Due to **weak governance, unrealistic planning, increasing costs, misalignment** with the stakeholders and **immature DLT implementation** resulted in the **project failure**.



Governance



Issue

Weak governance and oversight throughout the project cycle.



Evidence

- Governance and delivery weaknesses identified.
- Multiple project resets reflected poor project control.



Impact

- Delays and cost escalation.
- Loss of stakeholder confidence.



Key Takeaway

Strong governance and accountability are critical for project success.



Stakeholders



Issue

Stakeholder alignment and confidence deteriorated over time.



Evidence

- Industry concerns over cost, migration and transparency.
- Declining stakeholder trust.



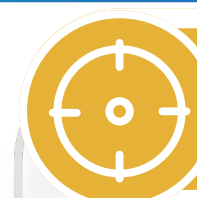
Impact

- Loss of stakeholder confidence.
- Greater resistance to implementation.



Key Takeaway

Early engagement and clear communication build stakeholder trust.



Scope



Issue

Complex scope and evolving requirements.



Evidence

- New DLT technology.
- Complex system integration.



Impact

- Higher costs and delays.
- Reduced delivery confidence.



Key Takeaway

Clear scope and change control prevent scope creep.



Schedule



Issue

Aggressive and unrealistic timelines led to repeated delays.



Evidence

- Delivery milestones repeatedly revised.
- Integration and testing exceeded planned timelines.



Impact

- Increased costs and resource strain.
- Reduced confidence in project delivery.



Key Takeaway

Realistic scheduling and regular re-estimation improve project delivery.



Finance



Issue

Significant cost overruns beyond initial estimates.



Evidence

- Investment increased significantly.
- AUD 245–255 million written off after project failure.



Impact

- Financial loss for ASX.
- Reduced investment capacity.



Key Takeaway

Strong financial controls and regular cost reviews reduce project risk.



Resource



Issue

Insufficient skills and coordination for a complex DLT project.



Evidence

- Limited DLT expertise.
- Complex vendor and team coordination.



Impact

- Increased rework.
- Delays and higher costs.



Key Takeaway

Skilled teams and strong vendor coordination support successful delivery.

Risk ID

Risk Title

Cause Description

Likelihood

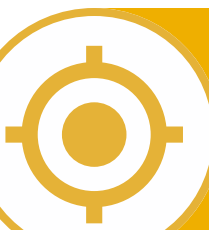
Impact



R1

**Governance
Failure**Weak oversight and
decision-making**High****High**

R2

**Schedule
Delays**Unrealistic timelines and
repeated plannings**High****High**

R3

**Scope
Creep**Expanding requirements
and system complexity**Medium****High**

R4

**Budget
Overrun**

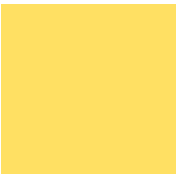
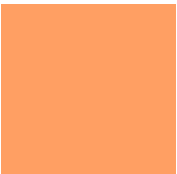
Escalating project costs

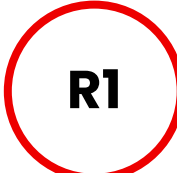
**High****High**

R5

**Resource
Constraints**Limited DLT expertise and
vendor coordination**High****Medium**

		IMPACT		
		Low	Medium	High
LIKELIHOOD	High		R5	R1 R2 R4
	Medium			R3
	Low			

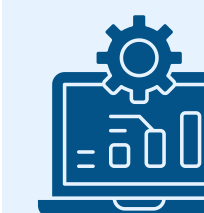
	Low Risk (Monitor)		Medium Risk (Monitor & Review)		High Risk (Mitigate)		Very High Risk (Critical - Act)
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	R1 Governance Failure		R2 Schedule Delays		R3 Scope Creep		R4 Budget Overrun		R5 Resource Constraints
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Critical Risks

R1, R2, and R3 are in the Very High Risk zone.



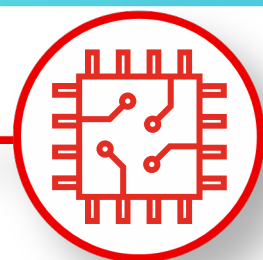
Key Contribution to Failure

The delays, escalation of cost, and loss of stakeholder trust were caused by these risks that had the highest likelihood and impact.



Overall Influence

The failure of the CHES Replacement Project was due to the combination of high likelihood and high impact of the risks.



1

Adopt DLT as the Core Platform

Effectiveness



✗ Ineffective

DLT technology was immature and unproven at scale

What Could Have Been Improved?



Better validation of technology maturity and feasibility.

Alternative Action



Pilot DLT in a limited environment before full implementation



2

Continue Despite Repeated Delays

Effectiveness



✗ Ineffective

Delays were allowed to escalate without decisive escalation

What Could Have Been Improved?



Earlier reassessment of project viability and delivery strategy.

Alternative Action



Implement stage-gate reviews with clear go/no-go decision points.



3

Expand Project Scope

Effectiveness



⚠ Partially Ineffective

Scope expansion increased complexity, cost and delivery risk.

What Could Have Been Improved?



Stronger scope definition and change control processes.

Alternative Action



Prioritise core functionality first; defer non-essential requirements.



4

Continue Large-Scale Investment

Effectiveness



✗ Ineffective

Escalating costs led to significant write-offs and loss of value.

What Could Have Been Improved?



Stronger financial oversight and cost control throughout.

Alternative Action



Release funding in phases based on milestone delivery and value realisation.



Key Takeaway

Earlier validation, stronger governance, stricter scope control and disciplined financial management could have significantly improved the project's outcome.

1 Major Lessons Learned

- Strong governance drives project success.
- Realistic planning reduces delivery risk.
- Early stakeholder engagement builds trust.
- Effective change control prevents scope creep.

2 What I Would Do Differently

- Validate technology readiness before implementation.
- Apply stage-gate reviews with go or no-go decisions.
- Escalate critical issues earlier.
- Strengthen communication with stakeholders.

3 Future Project Management Practices

- Adopt risk-based project planning.
- Maintain continuous stakeholder engagement.
- Conduct regular schedule and budget reviews.
- Implement independent governance and assurance.

4 Personal Reflection

This case study helped increase my understanding the importance of governance, stakeholder, communication, risk management and good decision-making can influence the project success. When leading complex projects that are technology related, it is important to balance the innovation and discipline in project management.

Strong governance, stakeholder engagement that is effective, disciplined planning, and good risk management are important for project's success.

References

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AI Declaration

ChatGPT was used to refine language, organize the content, and improve the wordings. The final work was reviewed and verified through the cited sources.